Hexagon Number Pyramids

Year 4 · Lesson Worksheet

Worked Examples

Easy: A 3-hexagon base row has the numbers 4, 7, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?

Answer: Row 2: 4+7=11, 7+5=12. Top row: 11+12=23.

Standard: A 4-hexagon base row has 3, 6, 2, 8. What is the top number?

Answer: Row 2: 3+6=9, 6+2=8, 2+8=10. Row 3: 9+8=17, 8+10=18. Top: 17+18=35.

Advanced: A 3-hexagon base row has 6, 9 and an unknown third number. The top hexagon is 40. Find the missing base number.

Answer: Let the missing number be x. Row 2: 6+9=15, 9+x. Top = 15+(9+x) = 24+x = 40, so x = 16.

Easy / Standard

(15 questions)

1. A 3-hexagon base row has the numbers 5, 6, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?
2. A 3-hexagon base row has the numbers 13, 8, 9. Each hexagon above is the sum of the two hexagons below it. What is the top number?
3. A 3-hexagon base row has the numbers 4, 3, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?
4. A 3-hexagon base row has the numbers 2, 8, 10. Each hexagon above is the sum of the two hexagons below it. What is the top number?
5. A 3-hexagon base row has the numbers 6, 14, 14. Each hexagon above is the sum of the two hexagons below it. What is the top number?
6. A 3-hexagon base row has the numbers 2, 5, 10. Each hexagon above is the sum of the two hexagons below it. What is the top number?
7. A 3-hexagon base row has the numbers 10, 7, 6. Each hexagon above is the sum of the two hexagons below it. What is the top number?
8. A 3-hexagon base row has the numbers 14, 4, 15. Each hexagon above is the sum of the two hexagons below it. What is the top number?
9. A 3-hexagon base row has the numbers 3, 6, 5. Each hexagon above is the sum of the two hexagons below it. What is the top number?

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10. A 3-hexagon base row has the numbers 2, 15, 12. Each hexagon above is the sum of the two hexagons below it. What is the top number?

11. A 3-hexagon base row has the numbers 14, 6, 14. Each hexagon above is the sum of the two hexagons below it. What is the top number?

12. A 3-hexagon base row has the numbers 6, 5, 4. Each hexagon above is the sum of the two hexagons below it. What is the top number?

13. A 3-hexagon base row has the numbers 6, 6, 12. Each hexagon above is the sum of the two hexagons below it. What is the top number?

14. A 3-hexagon base row has the numbers 15, 13, 15. Each hexagon above is the sum of the two hexagons below it. What is the top number?

15. A 3-hexagon base row has the numbers 15, 7, 3. Each hexagon above is the sum of the two hexagons below it. What is the top number?

Standard

(15 questions)

1. A 4-hexagon base row is 13, 15, ?, 10. The second row up reads 28, 34, 29 and the top hexagon is 125 (addition rule). What is the missing base number?

2. A 4-hexagon base row is 8, 10, ?, 11. The second row up reads 18, 28, 29 and the top hexagon is 103 (addition rule). What is the missing base number?

3. A 4-hexagon base row is 5, 20, ?, 3. The second row up reads 25, 32, 15 and the top hexagon is 104 (addition rule). What is the missing base number?

4. A 4-hexagon base row is 12, 12, ?, 9. The second row up reads 24, 31, 28 and the top hexagon is 114 (addition rule). What is the missing base number?

5. A 4-hexagon base row has 16, 16, 12, 16 (each hexagon above = sum of the two below). What is the top number? A) 116 B) 121 C) 100 D) 132

6. A 4-hexagon base row is 12, 11, ?, 5. The second row up reads 23, 15, 9 and the top hexagon is 62 (addition rule). What is the missing base number?

7. A 4-hexagon base row is 4, 17, ?, 19. The second row up reads 21, 28, 30 and the top hexagon is 107 (addition rule). What is the missing base number?

8. A 4-hexagon base row is 20, 18, ?, 7. The second row up reads 38, 31, 20 and the top hexagon is 120 (addition rule). What is the missing base number?

9. A 4-hexagon base row is 9, 5, ?, 9. The second row up reads 14, 21, 25 and the top hexagon is 81 (addition rule). What is the missing base number?

10. A 4-hexagon base row has 17, 11, 8, 14 (each hexagon above = sum of the two below). What is the top number? A) 77 B) 88 C) 93 D) 105

11. A 4-hexagon base row is 20, 9, ?, 6. The second row up reads 29, 22, 19 and the top hexagon is 92 (addition rule). What is the missing base number?

12. A 4-hexagon base row is 4, 10, ?, 10. The second row up reads 14, 21, 21 and the top hexagon is 77 (addition rule). What is the missing base number?

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13. A 4-hexagon base row is 6, 13, ?, 12. The second row up reads 19, 21, 20 and the top hexagon is 81 (addition rule). What is the missing base number?

14. A 4-hexagon base row is 17, 3, ?, 14. The second row up reads 20, 7, 18 and the top hexagon is 52 (addition rule). What is the missing base number?

15. A 4-hexagon base row has 5, 12, 13, 3 (each hexagon above = sum of the two below). What is the top number? A) 71 B) 92 C) 83 D) 88

Standard / Hard

(15 questions)

1. A 5-hexagon base row has 8, 24, 17, 23, 25. Each hexagon above is the sum of the two below. What is the single top number?

2. A 3-hexagon base row has 7, 14 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 79. Find the missing base number.

3. Build the full pyramid from base row 9, 17, 12 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

4. A 5-hexagon base row has 8, 12, 16, 23, 9. Each hexagon above is the sum of the two below. What is the single top number?

5. A 3-hexagon base row has 15, 5 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 63. Find the missing base number.

6. Build the full pyramid from base row 4, 17, 8 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

7. A 5-hexagon base row has 15, 15, 13, 22, 7. Each hexagon above is the sum of the two below. What is the single top number?

8. A 3-hexagon base row has 19, 11 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 67. Find the missing base number.

9. Build the full pyramid from base row 9, 6, 4 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

10. A 5-hexagon base row has 5, 5, 9, 23, 25. Each hexagon above is the sum of the two below. What is the single top number?

11. A 3-hexagon base row has 9, 6 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 74. Find the missing base number.

12. Build the full pyramid from base row 18, 10, 13 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

13. A 5-hexagon base row has 5, 7, 20, 13, 17. Each hexagon above is the sum of the two below. What is the single top number?

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14. A 3-hexagon base row has 11, 20 and an unknown third number. The top hexagon (sum of all three combined through the addition rule) is 52. Find the missing base number.

15. Build the full pyramid from base row 10, 17, 16 (addition rule). What is the sum of every number in the whole pyramid (all rows added together)?

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